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|---------------------------|-------------------------------|
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| <b>Section</b>            | B.com (Computer Applications) |
| <b>Hall ticket number</b> | 0225046035                    |

### Data Analytics with Tableau Assignment

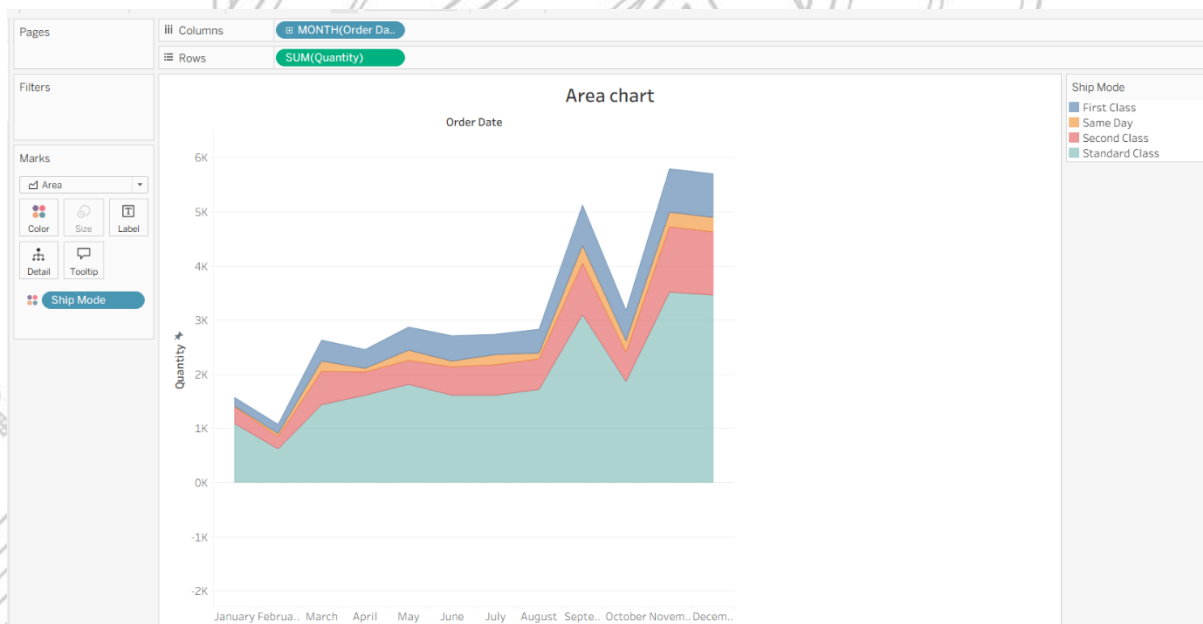
QUESTION: Prepare the following charts

#### Sheet - 01 – Area chart

##### Steps:

1. Drag **Order Date** to the **Columns** shelf and change the date level to **Month** by right-clicking it.
2. Drag **Quantity** to the **Rows** shelf.
3. Place **Ship Mode** on the **Color** option in the **Marks** card.
4. In the **Marks** card, open the mark type drop-down and select **Area** to create the area chart.

##### Output



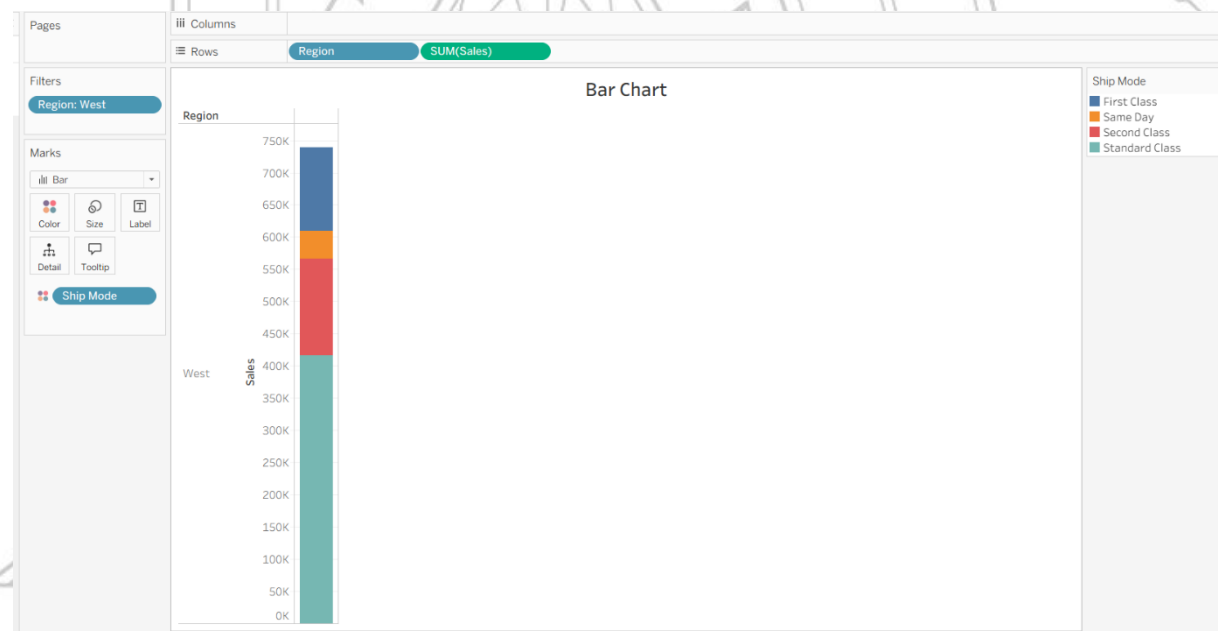
— ACHIEVERS —

## Sheet – 02 – Bar Chart

### Steps:

1. Drag **Sales** to the **Rows** shelf.
2. From the **Marks** card drop-down menu, select **Bar** as the chart type.
3. Drag **Ship Mode** to the **Color** option in the **Marks** card.
4. Drag **Region** to the **Rows** shelf and place it before **Sales**.
5. Drag **Region** to the **Filters** shelf and choose **West** to display only West region data.

### Output:



## Sheet – 03 – Box plot

### Steps:

1. Place **Sales** on the **Rows** shelf to display sales values.
2. Change the **Marks** type to **Bar** from the Marks card.
3. Drag **Ship Mode** to **Color** to distinguish each shipping method.
4. Add **Region** to the **Rows** shelf and position it before **Sales**.
5. Use **Region** as a filter and select **West** to view data for the West region only.

## Output:

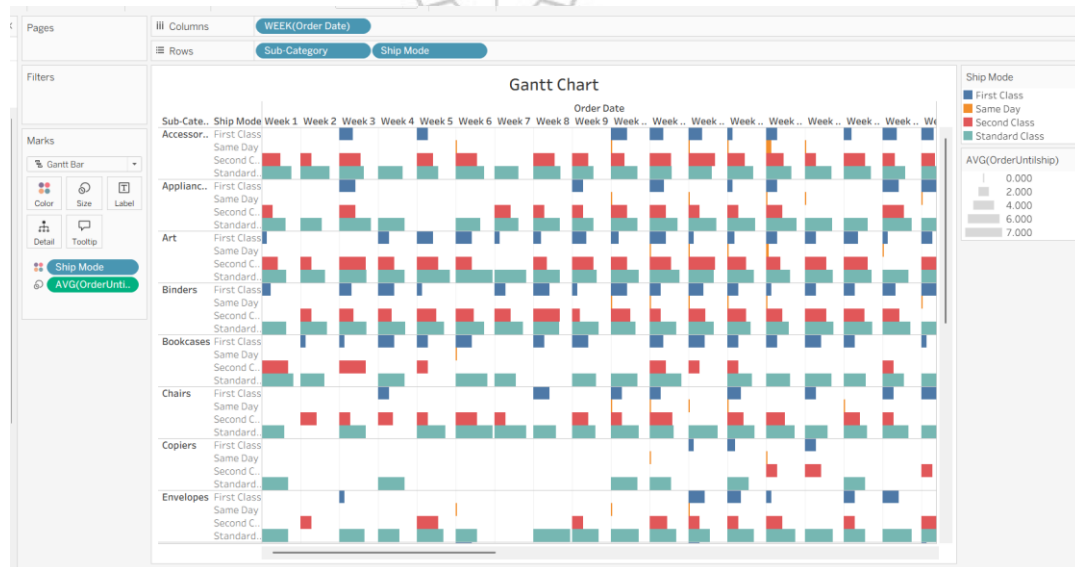


## Sheet – 04 – Gantt chart

### Steps:

1. Drag **Order Date** to the **Columns** shelf.
2. Open the **Order Date** drop-down menu and select **More → Week Number** to display data by week.
3. Drag **Sub-Category** to the **Rows** shelf, then place **Ship Mode** next to it.
4. Change the **Marks** type to **Gantt Bar** from the Marks card.
5. Create a calculated field named **OrderUntilShip** using the formula:  
DATEDIFF('day', [Order Date], [Ship Date])
6. Drag **OrderUntilShip** to **Size** on the Marks card.
7. Right-click **OrderUntilShip** on the Marks card and change its aggregation to **Average**.
8. Drag **Ship Mode** to **Color** on the Marks card to differentiate the Gantt bars by shipping method.

## Output:

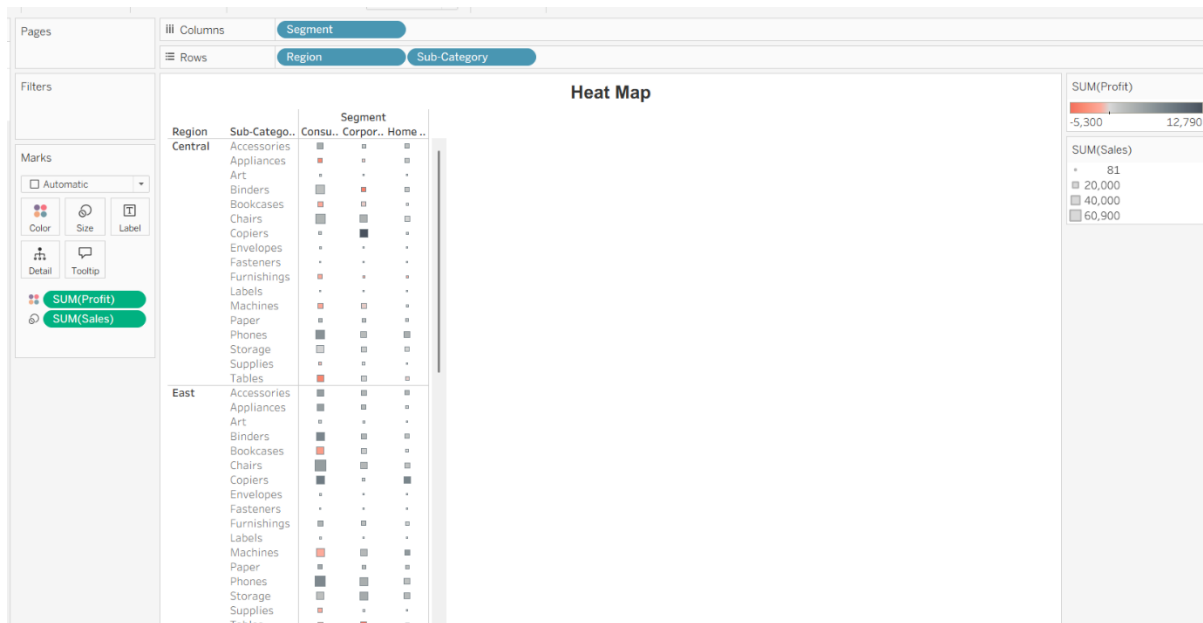


## Sheet – 05 – Heat Map

### Steps:

1. Drag **Segment** to the **Columns** shelf and place **Region** and **Sub-Category** on the **Rows** shelf.
2. Drag **Profit** to **Color** on the **Marks** card and choose the **Red-Black Diverging** color palette.
3. Drag **Sales** to **Size** on the **Marks** card and adjust the size slider to fit the boxes within the cells.
4. Click **Color** on the **Marks** card and set the **Border** color to **Medium Gray** for better visual clarity.

## Output:



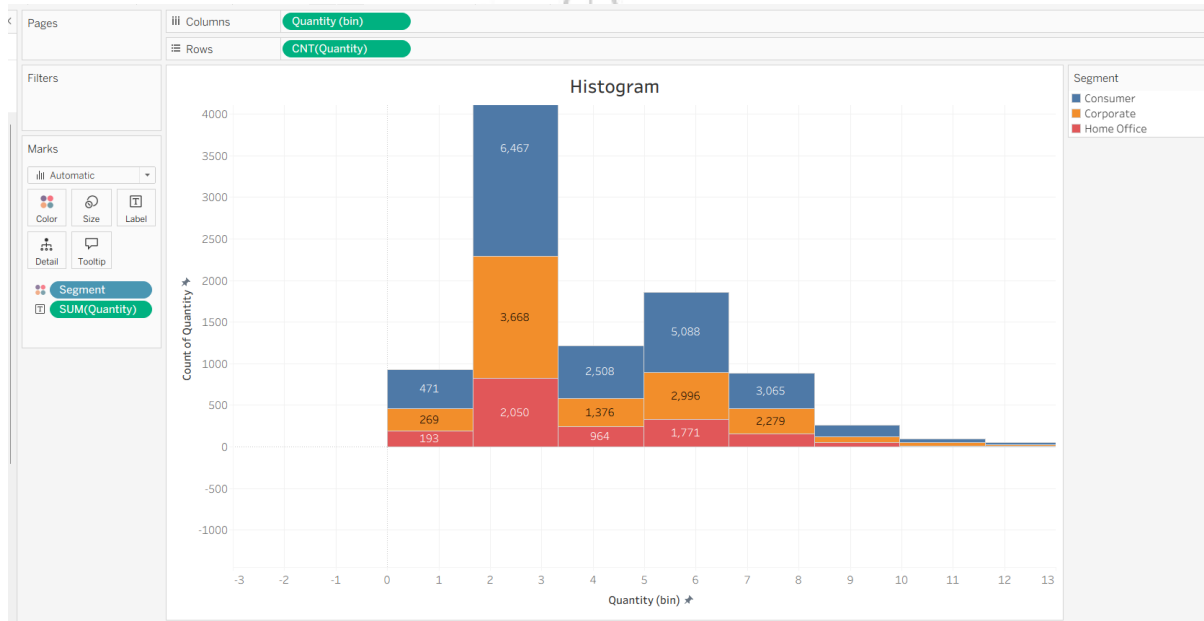
## Sheet – 06 – Histogram

### Steps:

1. Drag **Quantity** to the **Columns** shelf and select **Histogram** from the **Show Me** panel to automatically generate bins and frequency counts.
2. Drag **Segment** to **Color** on the **Marks** card to create a stacked histogram by segment.
3. Hold **Ctrl** and drag **CNT(Quantity)** from the **Rows** shelf to the **Label** option on the Marks card.
4. Right-click **CNT(Quantity)** on the **Marks** card, choose **Quick Table Calculation**, and select **Percent of Total** to display percentages.

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## Output:



## Sheet – 07 – Line Chart

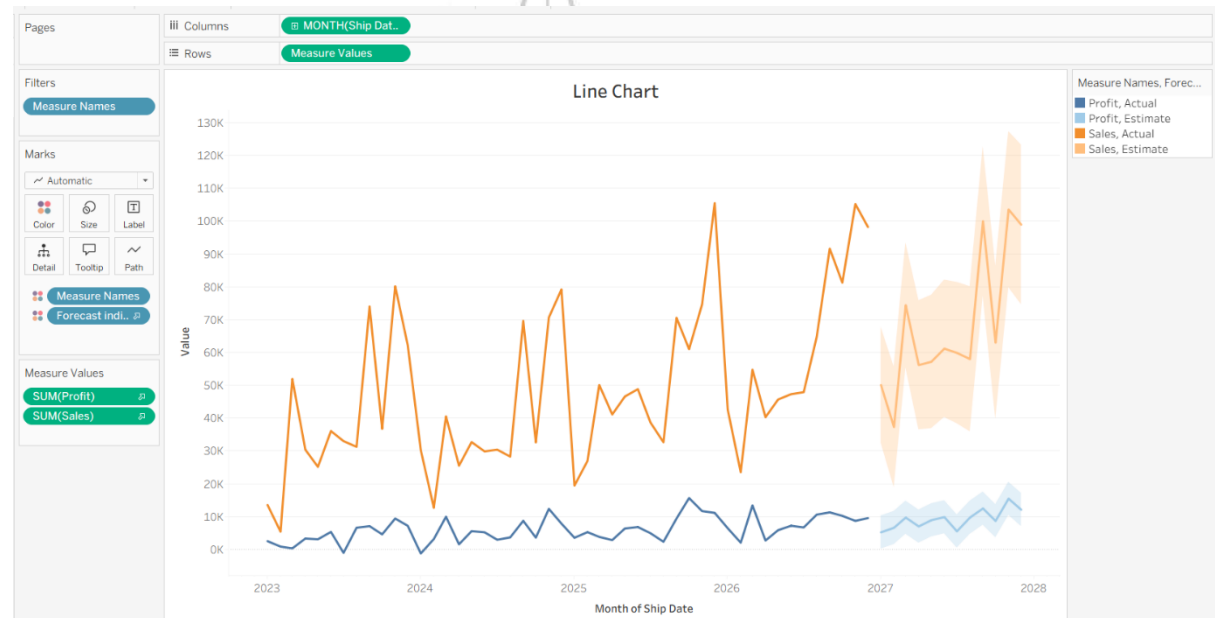
### Steps:

1. Drag **Ship Date** to the **Columns** shelf and change it to **Month (Continuous)** from the lower section of the date drop-down menu.
2. Drag **Sales** to the **Rows** shelf, then drag **Profit** directly onto the existing **Sales** axis to create a blended axis chart.
3. Ensure **Measure Names** is placed on the **Color** option in the **Marks** card to differentiate the **Sales** and **Profit** lines.
4. Open the **Analytics** pane, drag **Forecast** into the view, and drop it onto the chart to display predicted future trends.

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## Output:

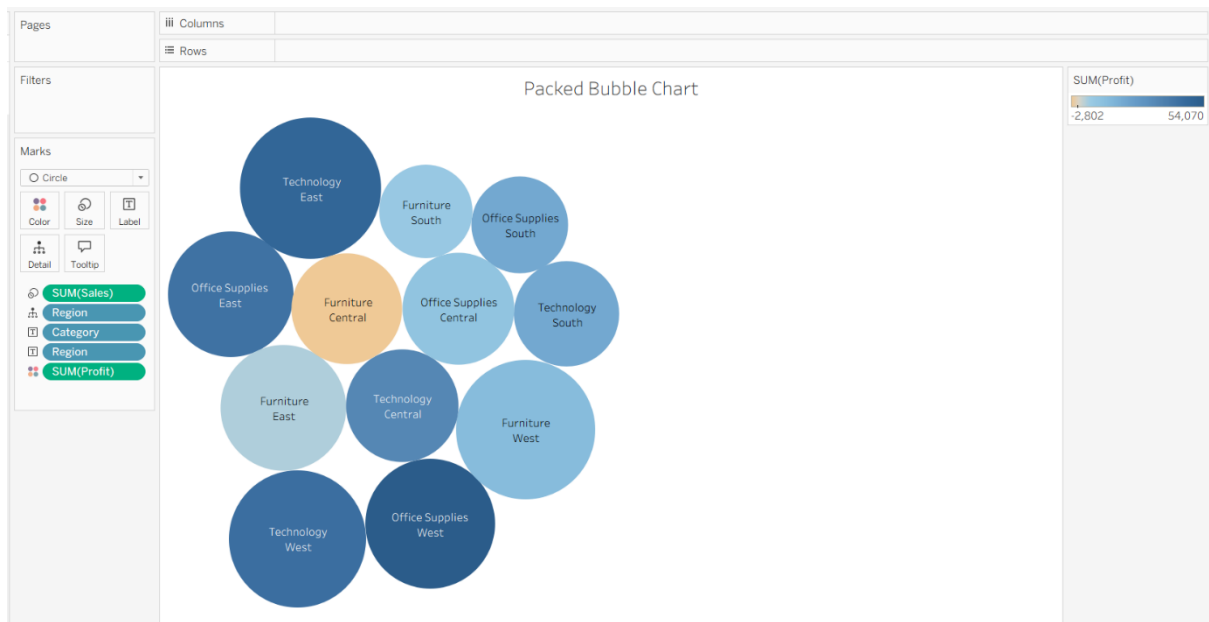


## Sheet – 08 – Packed Bubble Chart

### Steps:

1. Drag **Category** to the **Columns** shelf and **Region** to the **Rows** shelf, then select **Packed Bubbles** from the **Show Me** panel.
2. Drag **Sales** to **Size** on the **Marks** card to control the size of each bubble.
3. Drag **Profit** to **Color** on the **Marks** card to represent profitability using color variations.
4. Drag both **Category** and **Region** to **Label** on the **Marks** card so each bubble displays its corresponding data segment.

## Output:



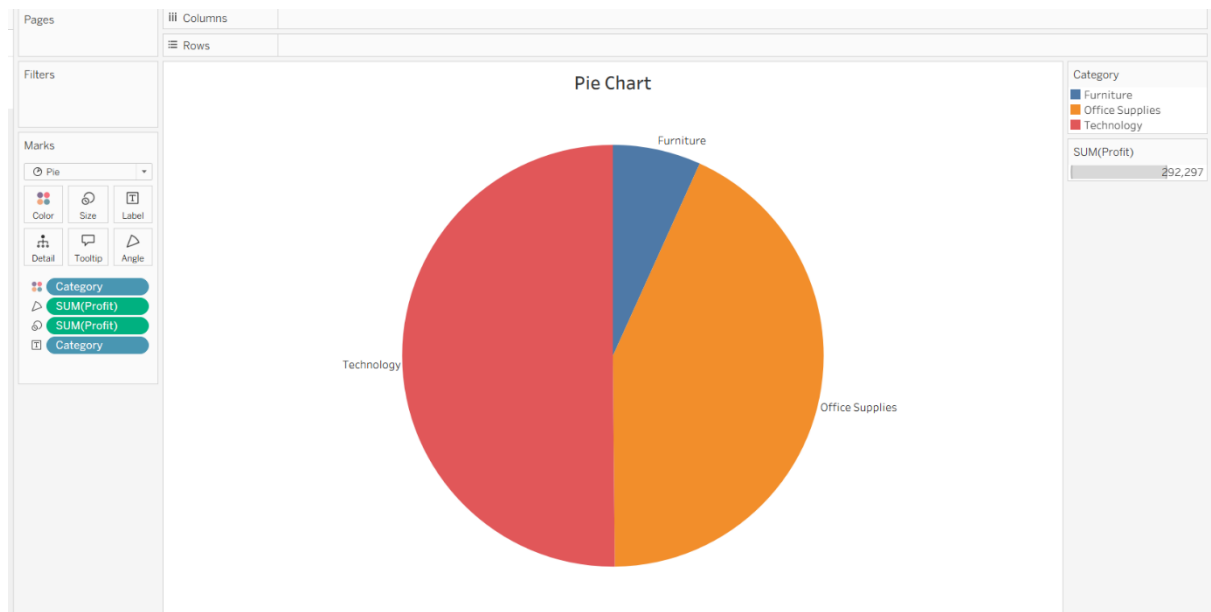
## Sheet – 09 – Pie Chart

### Steps:

1. Drag **Profit** to the **Columns** shelf and **Category** to the **Rows** shelf, then choose **Pie Chart** from the **Show Me** panel.
2. Place **Category** on the **Color** option in the **Marks** card to create separate slices for each category.
3. Drag **Profit** to the **Angle** mark to determine the size of each pie slice.
4. Drag **Category** to the **Label** mark so that each slice displays its category name.



## Output:



## Sheet – 10 – Scatter Plot

### Steps:

1. Drag **Sales** to the **Columns** shelf and **Profit** to the **Rows** shelf.
2. Drag **Category** to **Color** and **Region** to **Detail** on the **Marks** card to create multiple data points.
3. Open the **Analytics** pane, drag **Trend Line** into the view, and choose the **Linear** model.
4. Click **Shape** on the **Marks** card, select **Open Circle** shapes, and adjust the transparency to match the desired output style.

## Output:



## Sheet – 11 – Text Table with Percentages

### Steps:

1. Drag **Order Date** to the **Columns** shelf and place **Region** and **Sub-Category** on the **Rows** shelf.
2. Drag **Sales** to the **Text (ABC)** mark on the **Marks** card to populate the table with sales values.
3. Right-click **SUM(Sales)** on the **Marks** card and select **Quick Table Calculation** → **Percent of Total**.
4. Right-click **SUM(Sales)** again, choose **Compute Using**, and select **Pane (Down)** to calculate percentages within each region.

## Output:

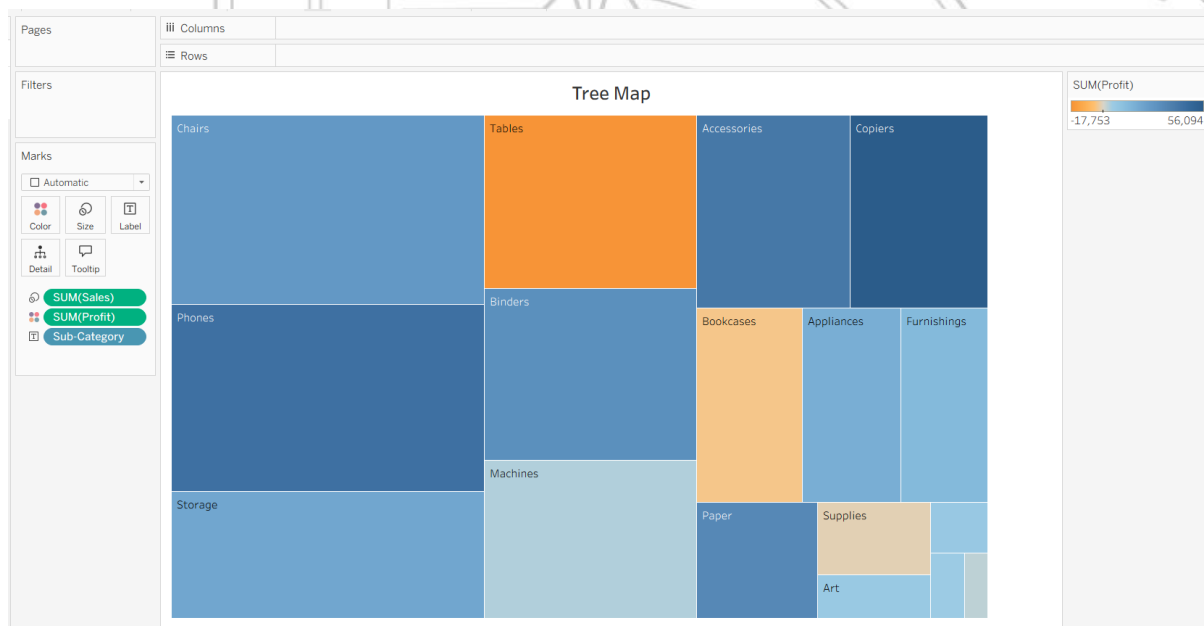
Text Table With Percentage

|         |              | Order Date |        |        |        |
|---------|--------------|------------|--------|--------|--------|
|         |              | 2023       | 2024   | 2025   | 2026   |
| Region  | Sub-Category |            |        |        |        |
| Central | Accessories  | 4.27%      | 7.58%  | 7.27%  | 7.39%  |
|         | Appliances   | 3.52%      | 4.84%  | 4.05%  | 6.04%  |
|         | Art          | 0.79%      | 1.10%  | 1.02%  | 1.55%  |
|         | Binders      | 15.28%     | 5.73%  | 10.19% | 14.31% |
|         | Bookcases    | 1.77%      | 8.07%  | 5.64%  | 3.81%  |
|         | Chairs       | 19.99%     | 17.41% | 15.74% | 16.15% |
|         | Copiers      | 3.15%      | 12.45% | 11.78% | 2.49%  |
|         | Envelopes    | 1.54%      | 0.85%  | 0.65%  | 0.81%  |
|         | Fasteners    | 0.12%      | 0.09%  | 0.17%  | 0.22%  |
|         | Furnishings  | 2.44%      | 2.46%  | 3.45%  | 3.44%  |
|         | Labels       | 1.01%      | 0.30%  | 0.34%  | 0.40%  |
|         | Machines     | 15.69%     | 1.80%  | 1.79%  | 4.05%  |
|         | Paper        | 2.26%      | 3.45%  | 3.61%  | 4.22%  |
|         | Phones       | 9.56%      | 18.82% | 13.39% | 15.70% |
| East    | Storage      | 10.68%     | 8.10%  | 8.63%  | 9.28%  |
|         | Supplies     | 0.42%      | 0.32%  | 2.89%  | 2.98%  |
|         | Tables       | 7.50%      | 6.67%  | 9.37%  | 7.16%  |
|         | Accessories  | 4.47%      | 11.40% | 3.41%  | 6.85%  |
|         | Appliances   | 4.53%      | 4.26%  | 5.22%  | 5.73%  |
|         | Art          | 1.01%      | 1.13%  | 1.11%  | 1.27%  |
|         | Binders      | 5.40%      | 9.18%  | 10.81% | 6.84%  |
|         | Bookcases    | 8.02%      | 12.51% | 3.30%  | 3.42%  |
|         | Chairs       | 16.89%     | 12.75% | 12.70% | 14.50% |
|         | Copiers      | 3.91%      | 3.16%  | 10.81% | 11.27% |
|         | Envelopes    | 0.99%      | 0.56%  | 0.71%  | 0.39%  |
|         | Fasteners    | 0.16%      | 0.14%  | 0.21%  | 0.13%  |
|         | Furnishings  | 3.32%      | 3.46%  | 5.43%  | 5.07%  |
|         | Labels       | 0.38%      | 0.33%  | 0.47%  | 0.42%  |
|         | Machines     | 10.81%     | 9.18%  | 13.80% | 5.46%  |
|         | Paper        | 3.25%      | 2.53%  | 2.71%  | 3.62%  |
|         | Phones       | 16.04%     | 14.50% | 11.76% | 16.79% |
|         | Storage      | 12.07%     | 8.48%  | 12.11% | 9.40%  |

## Sheet – 12 – Treemap

### Steps:

1. Drag **Sub-Category** to the **Columns** shelf and **Sales** to the **Rows** shelf, then select **Treemap** from the **Show Me** panel.
2. Place **Sub-Category** on the **Label** option in the **Marks** card to identify each rectangle.
3. Drag **Profit** to **Color** and **Sales** to **Size** on the **Marks** card to control the color and size of the treemap rectangles.
4. **Output:**

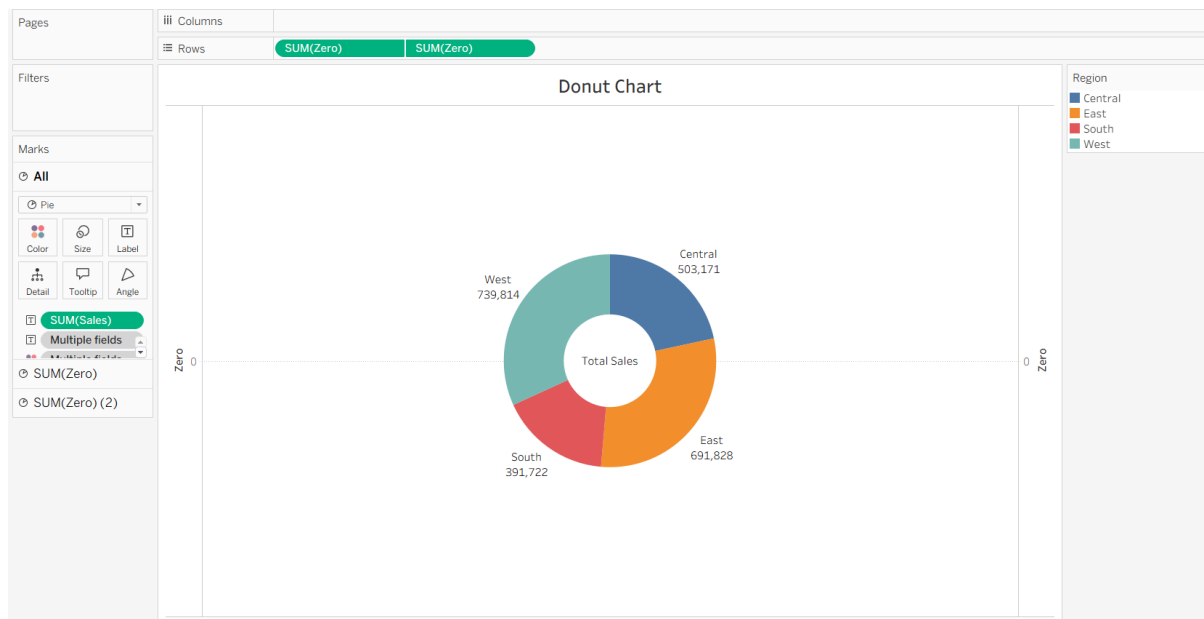


## Sheet – 13 – Donut chart

### Steps:

1. Create a calculated field named **Zero** with the formula 0, drag it twice to the **Rows** shelf, and set the **Marks** type to **Pie**.
2. On the first **SUM(Zero)** Marks card, drag **Region** to **Color** and **Sales** to **Angle**. On the second **SUM(Zero)** Marks card, remove all fields and set the color to **White**.
3. Right-click the second **SUM(Zero)** pill on the **Rows** shelf and select **Dual Axis**, then increase the size of the first pie chart to form the donut ring.
4. Select the second **Marks** card and drag **Sales** to **Label** to display the total value in the center of the donut chart.

## Output:

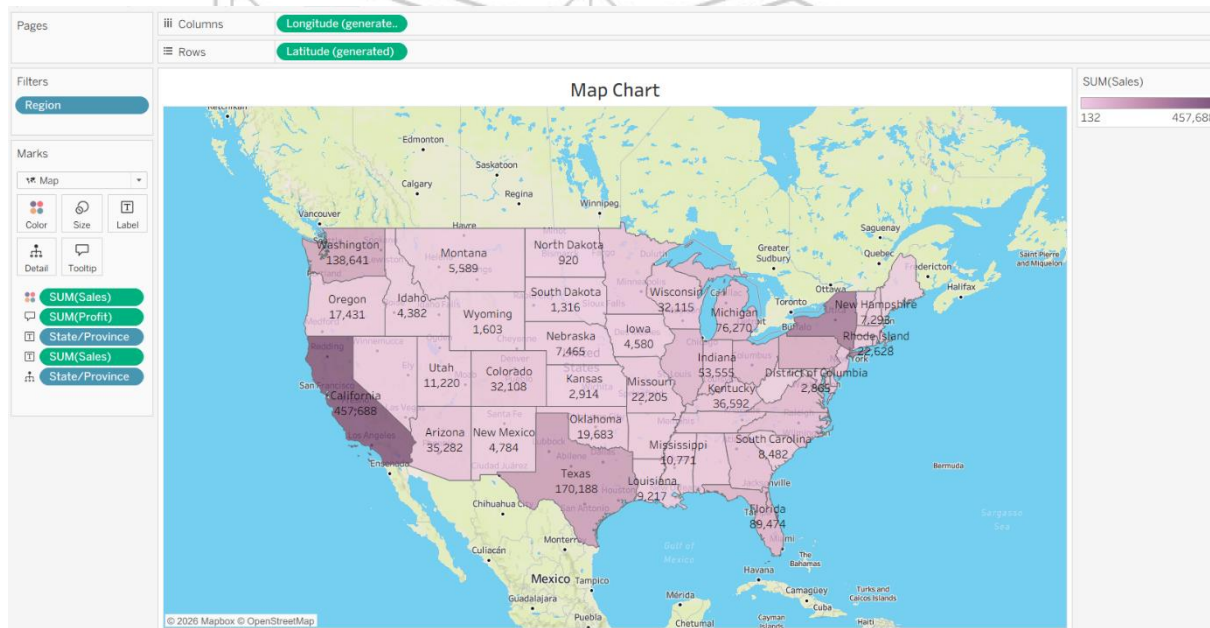


## Sheet – 14 – Map Chart

### Steps:

1. Double-click **State** (or drag it to the view) to automatically create a map using generated **Latitude** and **Longitude**.
2. Change the **Marks** type to **Map (Filled Map)** and drag **Sales** to **Color**, then apply a **Purple** color palette.
3. Drag **State** and **Sales** to **Label**, and place **Region** on the **Filters** shelf for interactive filtering.
4. Go to **Map → Map Layers**, select **Outdoors** as the background style, and set **Washout** to **70%** for improved readability.

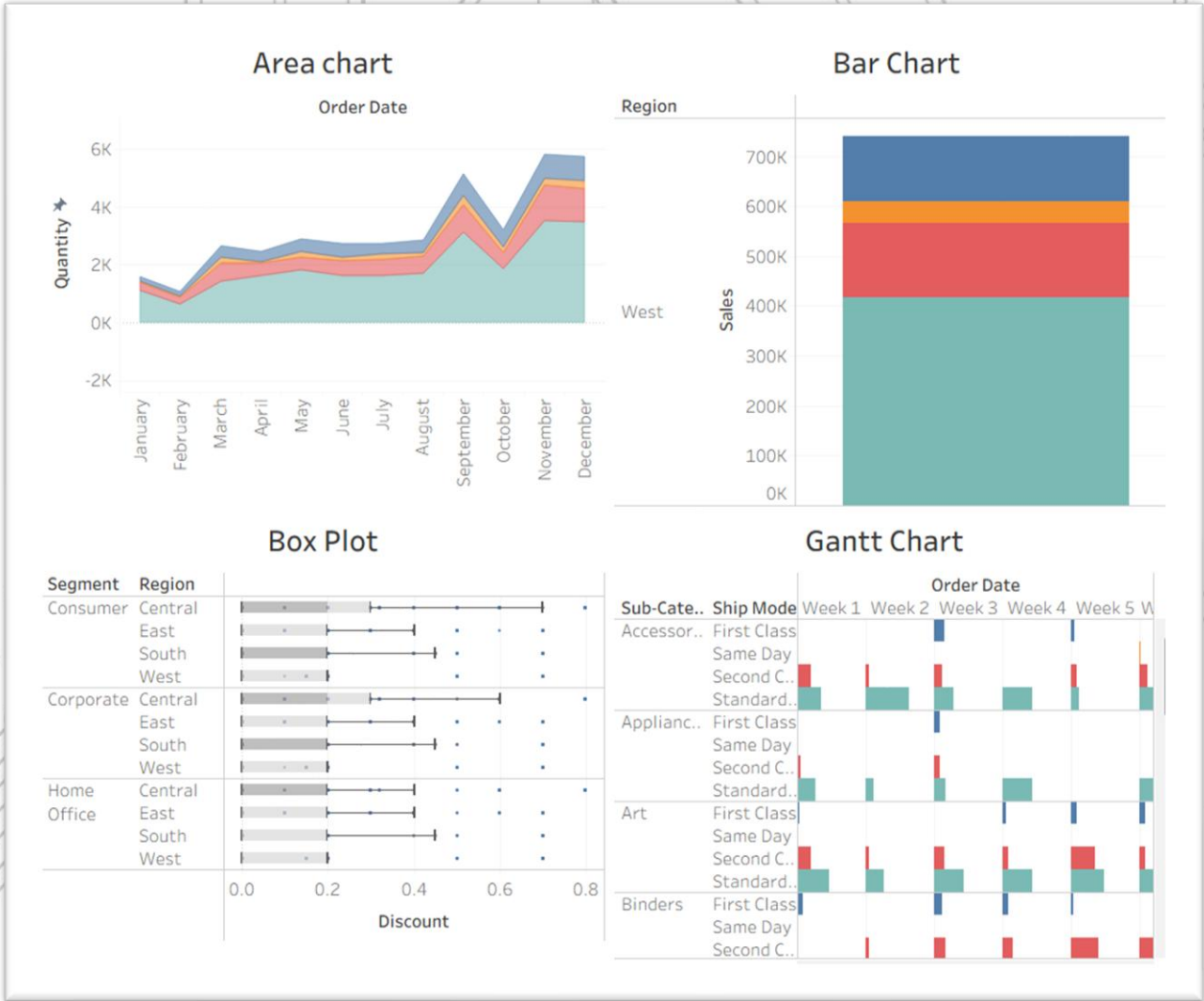
Output:



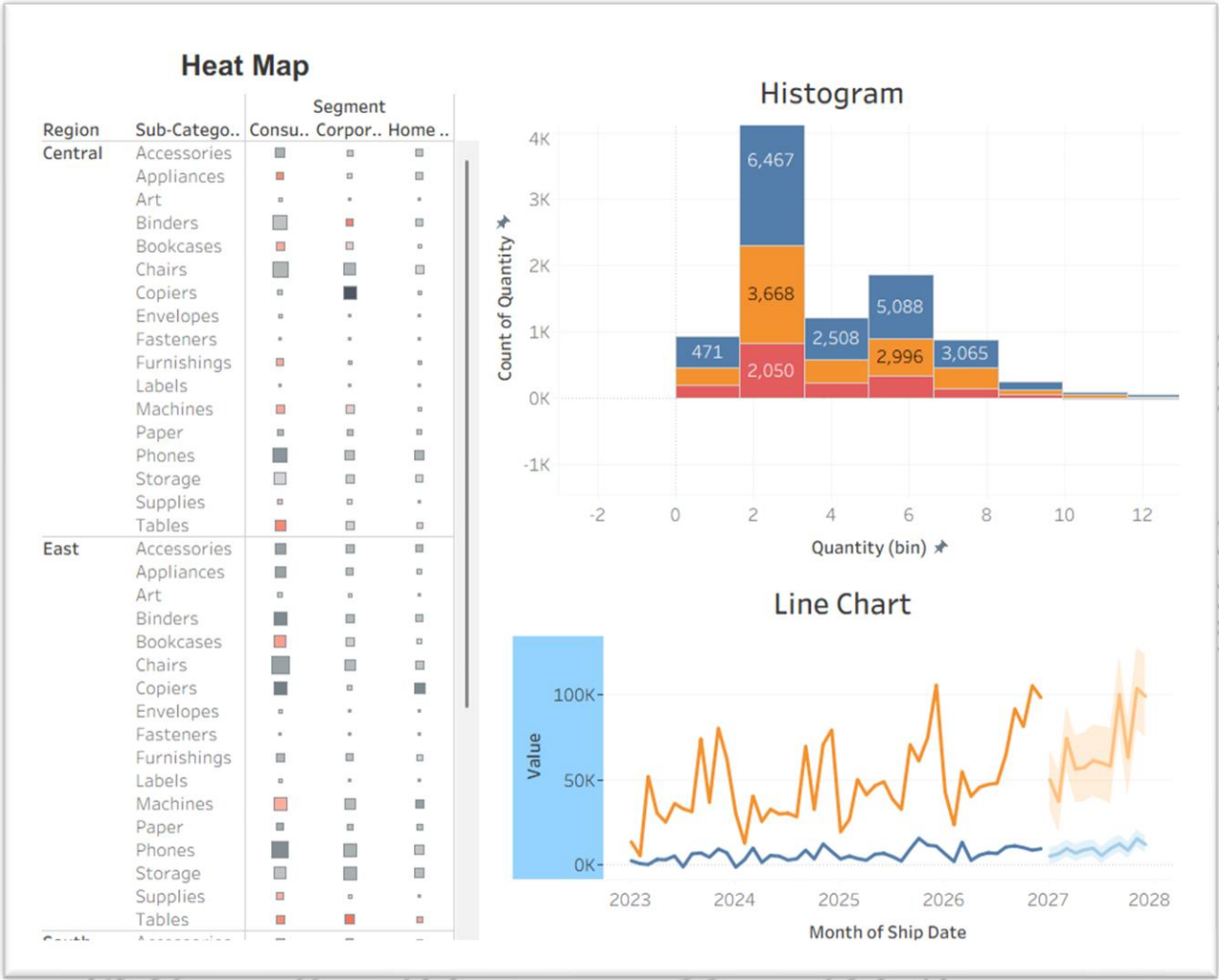
DELTA  
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DASHBOARDS:  
Dashboard – 01

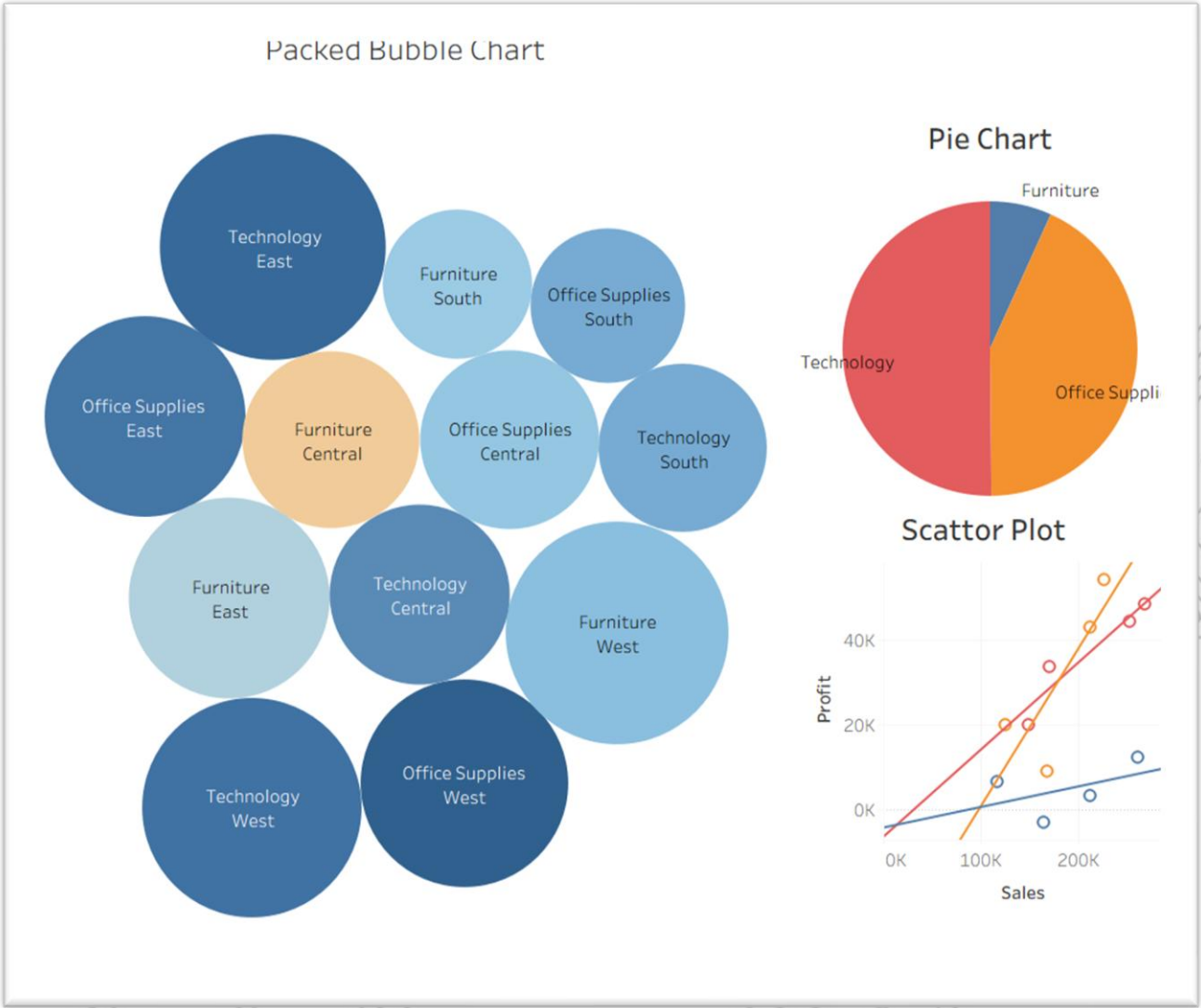


Dashboard – 02



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Dashboard – 03



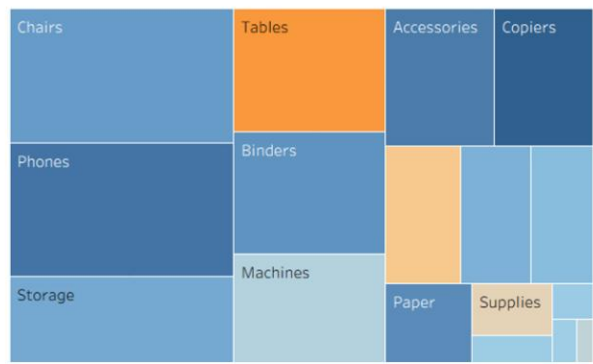
DELTA  
- ACHIEVERS -

Dashboard – 04

Text Table With Percentage

| Region  | Sub-Catego.. | Order Date |        |        |        |
|---------|--------------|------------|--------|--------|--------|
|         |              | 2023       | 2024   | 2025   | 2026   |
| Central | Accessories  | 4.27%      | 7.58%  | 7.27%  | 7.39%  |
|         | Appliances   | 3.52%      | 4.84%  | 4.05%  | 6.04%  |
|         | Art          | 0.79%      | 1.10%  | 1.02%  | 1.55%  |
|         | Binders      | 15.28%     | 5.73%  | 10.19% | 14.31% |
|         | Bookcases    | 1.77%      | 8.07%  | 5.64%  | 3.81%  |
|         | Chairs       | 19.99%     | 17.41% | 15.74% | 16.15% |
|         | Copiers      | 3.15%      | 12.45% | 11.78% | 2.49%  |
|         | Envelopes    | 1.54%      | 0.85%  | 0.65%  | 0.81%  |
|         | Fasteners    | 0.12%      | 0.09%  | 0.17%  | 0.22%  |
|         | Furnishings  | 2.44%      | 2.46%  | 3.45%  | 3.44%  |
|         | Labels       | 1.01%      | 0.30%  | 0.34%  | 0.40%  |
|         | Machines     | 15.69%     | 1.80%  | 1.79%  | 4.05%  |
|         | Paper        | 2.26%      | 3.45%  | 3.61%  | 4.22%  |
|         | Phones       | 9.56%      | 18.82% | 13.39% | 15.70% |
|         | Storage      | 10.68%     | 8.10%  | 8.63%  | 9.28%  |

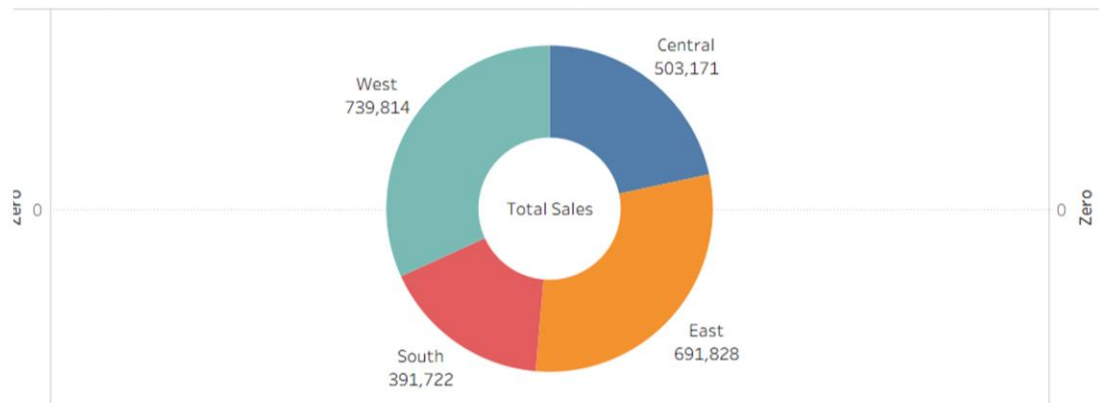
Tree Map



DELTA  
- ACHIEVERS -

## Dashboard – 05

Donut Chart



Map Chart



DELTA  
- ACHIEVERS -